

ECON 6018

TUTORIAL 5

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- Lesson Tutorial
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Today's Focus:

Implicit

Differentiation

Let's Recall:

$$y = x^2$$

Example:

ci) $y^2 = e^x$

cii) $y^2 + \ln y = x^2 + 2x$

Questions:

$$(i) \sqrt{ay^3} = (x^2 + 1)^{\frac{1}{3}}$$

$$(ii) e^y + e^x = 4$$

$$(iii) x^2 y^3 = 1$$

$$(iv) \frac{y^2}{x^3} + \ln x = 1$$

$$(v) ax^2 + b \ln xy^2 = c$$

$$(vi) e^{xy} + y = 1$$

$$(vii) g(x)^2 = (xy + b)^{2/3}$$

Answers:

$$(i) \sqrt[3]{ay^3} = (x^2 + 1)^{\frac{1}{3}}$$

$$a^{\frac{1}{3}} y^{\frac{3}{3}} = (x^2 + 1)^{\frac{1}{3}}$$

$$a^{\frac{1}{3}} \cdot \frac{dy}{dx} \cdot \frac{3}{3} y^{\frac{1}{3}} = \frac{1}{3} (x^2 + 1)^{-\frac{2}{3}} \cdot 2x$$

$$(ii) e^y + e^x = 4$$

$$e^y \frac{dy}{dx} + e^x = 0$$

$$(iii) x^2 y^3 = 1$$

$$2xy^3 + x^2 \frac{dy}{dx} \cdot 3y^2 = 0$$

$$(iv) \frac{y^2}{x^3} + \ln x = 1$$

$$\frac{2y \frac{dy}{dx} x^3 - y^2 3x^2}{x^6} + \frac{1}{x} = 0$$

$$(v) ax^2 + b \ln xy^2 = c$$

$$2ax + \frac{b \left(y^2 + x 2y \frac{dy}{dx} \right)}{x^2} = 0$$

$$(vi) e^{xy} + y = 1$$
$$(y + x \frac{dy}{dx}) e^{xy} + \frac{dy}{dx} = 0$$

$$(vii) g(x)^2 = (xy + b)^{2/3}$$

$$2g'(x) = \frac{2}{3} (xy + b)^{-\frac{1}{3}} \cdot (y + x \frac{dy}{dx})$$

Find the second order Derivatives:

(i) $x^3 y = 1$

(ii) $x^2 + y^2 = \ln y$

(iii) $\ln xy = 1$

Differentiate wrt. z

(i) $y^2 z = e^z + z^2$

(ii) $xz = ae^{xz}$

Answers:

(i) $x^3y = 1$

$$3x^2y + x^3 \frac{dy}{dx} = 0$$

$$3(2xy + x^2 \frac{dy}{dx}) + (3x^2 \frac{dy}{dx} + x^3 \frac{d^2y}{dx^2}) = 0$$

(ii) $x^2 + y^2 = \ln y$

$$2x + 2y \frac{dy}{dx} = \frac{1}{y} \cdot \frac{dy}{dx}$$

$$2 + 2 \cdot \frac{dy}{dx} + 2y \cdot \frac{d^2y}{dx^2} = -y^{-2} \cdot \frac{dy}{dx} + y^{-1} \cdot \frac{d^2y}{dx^2}$$

(iii) $\ln xy = 1$

$$\frac{y + x \frac{dy}{dx}}{xy} = 0$$

$$\frac{(\frac{dy}{dx} + (\frac{dy}{dx} + x \frac{d^2y}{dx^2}))(xy) - (y + x \frac{dy}{dx})^2}{(xy)^2}$$

$$(i) \quad y^2 z = e^z + z^2$$

$$2y \cdot \frac{dy}{dz} \cdot z + y^2 = e^z + 2z$$

$$(ii) \quad xz = ae^{xz}$$

$$\frac{dx}{dz} \cdot z + x = a \left(\frac{dx}{dz} \cdot z + x \right) e^{xz}$$